Structure, Swelling, and Molecular Weight Characterization of Polyacrylamide Hydrogels Synthesized Under Varied Formulation Conditions

Abstract

Polyacrylamide networks were synthesized under varied monomer and initiator conditions to quantify how polymerization kinetics, molecular weight, crosslink density, and environment control hydrogel structure and separation performance. Polymerization was monitored by temperature rise and viscosity growth, and an adiabatic energy balance was used to extract group kinetic constants, giving G = 0.047-0.095 min^{-1} for the adiabatic runs, in good agreement with literature values for APS/TEMED-initiated acrylamide. We combined adiabatic temperature monitoring, falling-ball and rotational viscometry, equilibrium swelling in water and saline, and SDS-PAGE separations over formulations spanning 3-6 wt % acrylamide, ~7.5-14.5 %T, and a range of temperatures and ionic strengths. Dilute-solution Ubbelohde viscometry and Huggins extrapolation yielded intrinsic viscosities $[\eta]$ = 1.6-5.8 dL g^{-1}, corresponding via Mark-Houwink to viscosity-average molecular weights M_v = 3.4×10^5-1.6×10^6 g mol^{-1}, with trends consistent with higher acrylamide concentration and lower effective initiator levels producing larger M_v.

Crosslinked PAAM hydrogels spanning ~7.5-14.5 %T and ~0.25-0.5 %C were characterized by short- (0-60 min) and long-term (7 day) swelling in water and saline media. All gels showed rapid initial uptake followed by slow elastic relaxation. Flory-Rehner analysis of equilibrium data gave molecular weights between crosslinks M_c = $(0.8\text{-}1.9)\times10^5$ g mol^{-1} and pore sizes ξ = 60-130 nm, with higher %T/%C and added salt reducing ξ and swelling.

SDS-PAGE on resolving gels of 8-12 %T and 2.9-4.0 %C demonstrated that intermediate formulations (10 %T, 2.9 %C) provided the best resolution for proteins in the 10-80 kDa range, whereas 12 %T gels largely blocked migration and none of the compositions resolved insulin (5.8 kDa). Together, these results establish a quantitative link from formulation and kinetics to network architecture and protein separation performance, and show that, despite run-to-run variability in student-prepared formulations, the trends are reproducible and consistent with classical polymer-physics and electrophoresis theory.

Introduction

Hydrogels are soft, water-swollen polymer networks whose high water content, tunable elasticity, and controllable permeability make them natural bridges between synthetic materials and biological tissues. Polyacrylamide hydrogels are especially useful as a model platform because network architecture (crosslink density) and porosity can be tuned precisely by formulation. Understanding how monomer concentration, crosslinker fraction, temperature, and ionic strength govern polymerization kinetics, molecular weight, swelling, and transport is essential for designing gels for biomolecular separations, drug delivery, and soft-material engineering.

This work quantifies how synthesis parameters map to functional properties through three linked components:(i) kinetics of free-radical polymerization of acrylamide and the viscosity-molecular-weight relationship in solution; (ii) network formation and equilibrium swelling of crosslinked hydrogels, analyzed with Flory-Rehner theory to extract the molecular weight between crosslinks (M_c) and pore size (ξ); and (iii) application to biomolecular separations by SDS-PAGE, where gel pore size controls protein resolution.

Our objective is to establish a predictive link from formulation (%T, %C, initiator) to network structure (M_c, ξ) to separation performance (resolution vs. size). In the Background that follows, we lay out the kinetic, rheological, and thermodynamic frameworks that motivate our experimental choices and enable quantitative interpretation of the results.

Background

Free-Radical Polymerization and Molecular Weight Characterization

Polyacrylamide (PAAM) forms via free-radical polymerization of acrylamide in water, proceeding through the familiar triad of initiation, propagation, and termination. In our system, initiation arises from ammonium persulfate (APS) activated by N,N,N′,N′-tetramethylethylenediamine (TEMED), generating radicals that add acrylamide monomer during propagation; chain growth ultimately ceases by radical recombination or disproportionation. The reaction is sufficiently exothermic that conversion can be inferred from the temperature-time trace under near-adiabatic conditions. Under typical conditions, the rate profile is expected to pass through three regimes: an initial induction period while dissolved oxygen is consumed and the radical population builds, a rapid auto-accelerating propagation regime associated with the Trommsdorff gel effect, and a final plateau as monomer is depleted and termination dominates. Increasing initiator (APS/TEMED) generally shortens the induction period and increases the early-time rate (larger G), but also creates more growing chains, leading to lower molecular weights at fixed conversion. In contrast, increasing monomer concentration boosts both the propagation rate and the viscosity build, often producing higher molecular weights because chain growth is favored over termination. In the early regime, a lumped kinetic "group constant" (G), captures the effective rate, with conversion approximated by $X_m(t) = 1 - e^{-Gt}$. This compact parameterization allows direct comparison across formulations.

To relate molecular architecture to macroscopic flow, we pair dilute-solution viscometry with Mark-Houwink analysis. The intrinsic viscosity $[\eta]$ is obtained from Ubbelohde capillary measurements by converting flow times to relative viscosity $\eta_{rel} = \frac{t}{t_0}$, then forming $\eta_{sp} = \eta_{rel} - 1$ and extrapolating either the reduced viscosity $\frac{\eta_{sp}}{c}$ or the inherent viscosity $\frac{ln(\eta_{rel})}{c}$ to $c \to 0$. In the Ubbelohde viscometer, gravity drives laminar flow through a narrow capillary, and the viscosity is inferred from the time required for the liquid meniscus to pass between two marks, assuming Poiseuille flow. With $[\eta]$ in hand, the viscosity-average molecular weight follows from the Mark-Houwink correlation $[\eta] = KM^a$ using PAAM constants of K = 1.21 × 10^{-4} dL g^{-1} and a = 0.746 at 25 °C in 0.5 M NaCl (Wagner et al.). Because solution behavior is often Newtonian only in the dilute limit, we complement capillary data with a rotational Bohlin coaxial Couette viscometer to characterize shear-rate dependence in semi-dilute samples. In the coaxial Couette geometry, the inner cylinder experiences a torque proportional to the viscous shear stress, so combining the measured torque with the rotation rate yields the apparent viscosity as a function of shear rate. A Newtonian fluid has a viscosity that is independent of shear rate, whereas non-Newtonian fluids, such as many polymer solutions, show a shear-dependent viscosity. There, many polymer solutions exhibit shear-thinning well described by the power-law relation $\eta = K\dot{\gamma}^{n-1}$, with $n < 1$. Together, these methods connect kinetic conditions to molecular weight and to rheology in the processing-relevant regime.

Crosslinked Polyacrylamide Gels

Crosslinking transforms linear PAAM into a water-swollen elastic network. During copolymerization with BIS, permanent covalent junctions form between chains; upon reaching a percolation threshold, viscosity rises sharply and a gel point is observed, marking the onset of solid-like response. Network architecture is conveniently parameterized by the total monomer concentration %T (w/v acrylamide + BIS) and the crosslinker fraction %C (BIS/(acrylamide + BIS) × 100%). Increasing %T and/or %C reduces the molecular weight between crosslinks M_c and the pore size ξ, yielding denser networks with smaller pores and higher modulus.

Swelling provides a thermodynamic window into this structure. At equilibrium, solvent uptake reflects a balance between the entropy of mixing and the elastic penalty for network stretching. The Flory-Rehner framework links the equilibrium polymer volume fraction φ_e to M_c, the solvent interaction parameter χ, and the solvent molar volume V_1; solving for M_c from measured φ allows estimation of ξ from PAAM chain statistics. For PAAM-water, χ ~ 0.4-0.5 indicates a good solvent, consistent with substantial swelling at low crosslink densities. Environmental variables modulate this balance: higher temperature generally increases chain mobility and favors swelling, while added salt (e.g., PBS or PBS + 5 M NaCl) screens charges and reduces the osmotic driving force, suppressing swelling. These dependencies directly anticipate transport behavior in sieving applications.

Sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE)
SDS-PAGE operationalizes this structure-transport link for protein separations. Sodium dodecyl sulfate (SDS) binds proteins at roughly one SDS per two amino acids, denaturing tertiary/quaternary structure and imposing a near-uniform charge-to-mass ratio. Under an applied field, migration then depends predominantly on hydrodynamic size relative to the gel pore. A discontinuous system comprising a low-%T stacking gel and a higher-%T resolving gel sharpens bands and effects size-based sieving, respectively. Buffer composition and pH maintain stable mobility and control Joule heating; voltages on the order of 200 V typically balance resolution with thermal management. After electrophoresis, Coomassie staining provides quantitative visualization; plotting the logarithm of molecular weight versus normalized travel distance yields a calibration used to estimate apparent protein sizes. Because %T and %C tune ξ, they define the effective resolution window: higher %T/%C improves discrimination among small proteins but can exclude very large species, whereas lower %T increases accessibility at the expense of fine resolution.

In summary, free-radical kinetics, dilute-solution viscometry, and rheology establish how synthesis variables set molecular weight and flow; Flory-Rehner analysis translates composition into M_c and ξ; and SDS-PAGE leverages this pore to deliver size-selective transport. This integrated framework underpins our experimental design and the interpretation of results that follow.

Experimental Methods

General Consideration and Safety
All experiments were conducted following institutional laboratory safety guidelines. Acrylamide monomer is a neurotoxin, APS is a strong oxidizer, and TEMED is a volatile irritant; therefore, all monomer handling, solution preparation, and gel casting were performed in a certified fume hood while wearing nitrile gloves, goggles, and a lab coat. Deoxygenation steps involving nitrogen sparging were carried out under gentle flow to prevent splashing. Waste containing unreacted acrylamide of TEMED was collected in labeled containers for hazardous disposal.

Once polymerization was complete, gels were considered non-toxic and handled on the benchtop with gloves until fully dried.

Synthesis and Characterization of Linear Polyacrylamide

(i) Preparation of Linear Polyacrylamide

Linear polyacrylamide was synthesized through free-radical polymerization of acrylamide in aqueous media initiated by APS and catalyzed by TEMED. The reaction compositions were designed to explore the influence of both monomer concentration and initiator level on polymerization kinetics and molecular weight. These values were selected from literature ranges that provide measurable viscosity changes while avoiding premature crosslinking or phase separation during polymerization.

A total of five formulations were prepared (Table 1), varying in acrylamide concentration (3-5 wt/vol%) and initiator ratio (0.04-0.08 g APS and 20-40 µL TEMED per 100 mL). Each solution was first prepared by dissolving the appropriate amount of acrylamide and APS in deionized water within a 300 mL Erlenmeyer flask. The solution was sparged with nitrogen for ten minutes to remove dissolved oxygen, which can inhibit free-radical initiation. Immediately following sparging, TEMED was introduced to initiate polymerization, marking the reaction start time $t = 0$.

Table 1. *Reaction compositions for linear polyacrylamide synthesis (per 100 mL solution).*

Team	Monomer	Initiator
1	"intermediate" AAM 4% wt vol	"high" 0.08 g APS, 40 uL TEMED
2	"intermediate" AAM 4% wt vol	"low" 0.04 g APS, 20 uL TEMED
3	"low" AAM 3% wt vol	"intermediate" 0.06 g APS, 30 uL TEMED
4	"high" AAM 5% wt vol	"intermediate" 0.06 g APS, 30 uL TEMED
5	"intermediate" AAM 4% wt vol	"intermediate" 0.06 g APS, 30 uL TEMED

(ii) Kinetic Monitoring

Polymerization progress was monitored through simultaneous measurements of temperature rise and viscosity change, each probing a different aspect of the reaction progression. To track the exothermic temperature profile, approximately 40 mL of the reaction mixture was transferred into an insulated Falcon tube immediately after initiation. A thermometer was inserted through Parafilm to minimize heat loss and contamination, and temperature readings were recorded every minute until a clear plateau was reached, which indicated that monomer conversion was nearly complete.

In parallel, viscosity evolution was followed using a simple falling-ball method. About 15 mL of the reacting mixture was poured into a separate tube, and a small glass bead was gently placed at the surface. The time required for the bead to traverse a fixed distance was measured every three minutes. The one-minute interval for temperature measurements and the three-minute interval for falling-ball measurements were selected to capture the onset of autoacceleration while avoiding heat-transfer artifacts or disturbances to the developing gel structure.

Because the propagation step is strongly exothermic, the temperature data provide a direct route to estimating conversion. Under near-adiabatic conditions, the conversion of acrylamide can be calculated using an energy balance,

$$\Delta H_r X_m = \rho C_p \Delta T,$$

where ΔH_r is the heat of polymerization (~80 kJ mol^{-1} monomer), X_m is the fractional monomer conversion, ρC_p the volumetric heat capacity (~4.2 kJ L^{-1} K^{-1} for aqueous solution), and ΔT the observed temperature rise.

(iii) Dilute-Solution Viscometry (Ubbelohde Method)
After polymerization, the intrinsic viscosity of the resulting polyacrylamide was determined using an Ubbelohde capillary viscometer to correlate molecular weight through the Mark-Houwink relation. The solutions were first diluted using 0.5 M NaCl containing 0.1 mM NaOH to prepare a four-point concentration series at 0.0010, 0.0015, 0.0020, and 0.0025 g mL^{-1}. The solvent flow time (t_0) using the dilution solution was recorded, followed by time measurements (t) of the diluted polymer solution by loading a 15 mL of solution into the viscometer, pulling the liquid above the upper mark using a syringe, and timing its fall between two marks as gravity drives laminar flow between marks. Flow time (t) was converted to relative viscosity $\eta_{rel} = \frac{t}{t_0}$, and reduced and inherent viscosities were calculated as $\frac{\eta_{sp}}{c}$ and $\frac{ln(\eta_{rel})}{c}$, respectively. Extrapolation of these quantities to zero concentration yielded intrinsic viscosity $[\eta]$. The viscosity-average molecular weight (M_v) was then obtained using the Mark-Houwink correlation $[\eta] = KM^a$ and parameters for PAAM in 0.5 M NaCl at 25 °C (K = 1.21 × 10^{-4} dL g^{-1}, a = 0.746).

Between measurements, the viscometer was rinsed thoroughly with solvent and dried with filtered air to eliminate cross-contamination. All measurements were repeated at least twice if possible.

(iv) Rotational Rheometry (Bohlin Coaxial Couette)
To assess flow behavior at higher concentrations, viscosity was also measured using a Bohlin coaxial Couette viscometer. This technique allowed quantification of shear-dependent viscosity and the extraction of the flow behavior index (n) from the power-law relation $\eta = K\gamma^{\cdot n-1}$. Apparent viscosity was recorded over a shear rate range of approximately 10-100 s^{-1} at 25 °C. Torque readings were maintained within ~10-90% of full scale, and equilibrium values were recorded after stabilization at each shear rate. The instrument gave us full-scale (FS) viscosity, and we calculated apparent viscosity from that using the formula:

$$\mu_{cP} = \left(\frac{T\%}{100}\right) \cdot \mu_{FS,\,cP}$$

Then the shear rate (γ) was estimated using shear-rate constant (SRC):

$$\gamma = SRC \cdot N_{rpm}, \qquad SRC = SRC_{RV6} \cdot \frac{D_{spindle}}{D_{RV6}},$$

where $SRC_{RV6} = 0.238\ \frac{1}{s \cdot rpm}$ and $D_{RV6} = 14.6\ mm.$ $D_{spindle}$ was different for each spindle type.

The use of both viscometers provided complementary insight: the Ubbelohde method captured intrinsic viscosity in the Newtonian, dilute regime, while the Couette system revealed non-Newtonian shear-thinning behavior in semi-dilute polymer solutions.

Crosslinked Polyacrylamide Hydrogel Synthesis and Swelling
(i) Synthesis of Crosslinked Polyacrylamide Hydrogels

Crosslinked polyacrylamide hydrogels were synthesized by free-radical copolymerization of acrylamide (AAM) and BIS in phosphate-buffered saline (PBS, 1x) using APS and TEMED as the redox initiator system. The synthesis was designed to investigate how variations in total monomer concentration (%T) and crosslinker ratio (%C) influence network density and swelling behavior.

Four formulations were prepared to span a representative range of gel compositions from soft, low-crosslink networks to dense, highly crosslinked matrices (Table 2). Stock solutions of 40 wt % AAM + 2 % BIS in PBS, 40 wt % AAM in PBS, 2 % BIS in PBS, and PBS alone were combined volumetrically to achieve the desired %T and %C values. PBS was used throughout as a buffered solvent to maintain near-neutral pH and suppress hydrolysis of amide groups to carboxylates during polymerization.

Table 2. *Composition of PAAM hydrogel formulations used by Teams 1-4.*
%T = total monomer concentration (wt./vol.); %C = BIS fraction relative to total monomer mass.

Team	Monomer %T	Crosslinker %C
1	"intermediate" 9.5%	"high" 0.5%
2	"intermediate" 9.75%	"low" 0.25%
3	"high" 14.5%	"intermediate" 0.5%
4	"low" 7.5%	"intermediate" 0.25%

For each 50 mL batch, the required volumes of stock solutions were mixed in a 250 mL Erlenmeyer flask. APS (0.05 g per 50 mL) was added and dissolved completely, and the solution was sparged with nitrogen for 5-10 min to remove dissolved oxygen, which can quench radical formation. Polymerization was initiated by adding 25 µL TEMED per 50 mL, and the mixture was gently swirled to ensure uniform catalyst dispersion. The solution was immediately poured into Petri dishes, covered to exclude air, and allowed to polymerize at room temperature. A small aliquot was retained in a sealed vial to monitor viscosity increase and confirm gelation. Polymerization typically reached completion within 30-60 min, producing transparent, elastic gels suitable for swelling analysis.

(ii) Swelling of Hydrogels
Swelling behavior was examined to quantify the solvent uptake and network porosity of each hydrogel formulation. After polymerization, gels were equilibrated in PBS for 1 hour to remove unreacted monomers, then cut into circular disks (8-10 mm diameter, 1-3 mm thickness) using biopsy punches. The initial (as-prepared) mass and dimensions were measured using ruler and mass balance. Dry reference values were obtained by air-drying separate disks to constant mass at ambient conditions.

Each hydrogel formulation (Table 2) was subjected to five environmental conditions designed to isolate entropic, osmotic, and thermodynamic effects on swelling: (A) DI water (22 °C), (B) DI water (9 °C), (C) DI water (58 °C), (D) PBS + 5 M NaCl, and (E) PBS. These conditions span physiologically relevant temperature and ionic strength ranges, enabling evaluation of solvent-polymer interactions and thermal sensitivity. For each formulation, four replicate disks were prepared. One served as the as-prepared/dry reference, while the remaining three were immersed in the designated swelling medium in labeled scintillation vials. At scheduled intervals (20, 40, 60 min and 60 min-7 days), a disk was gently blotted with Kimwipes to remove surface solvent, then weighed and measured. Equilibrium was verified when successive measurements showed < 2 % change in both mass and geometry. In addition to quantitative mass and

dimensional measurements, qualitative physical properties such as transparency, elasticity, and brittleness were noted for each formulation to correlate perceived stiffness with %T and %C.

From these measurements, the volumetric swelling ratio was determined as

$$Q_v = \frac{V_{sw}}{V_{dry}} \text{ where } V = \pi\left(\frac{d}{2}\right)^2 h.$$

The polymer volume fraction in the swollen state was then obtained as

$$\varphi_e = \frac{V_{polymer}}{V_{sw}} \sim \frac{m_{dry}/\rho_p}{V_{sw}}.$$

This value provides a direct experimental measure of network dilution, which is the key variable in the Flory-Rehner framework.

(iii) Flory-Rehner Analysis

The equilibrium polymer volume fraction φ_e was used to calculate the molecular weight between crosslinks $\bar{M}_c$ following the modified Flory-Rehner relation appropriate for gels synthesized in solvent:

$$\frac{1}{\bar{M}_c} = \frac{2}{\bar{M}_n} - \frac{\left(\bar{v}/V_1\right)\left[\ln(1-\varphi_e) + \varphi_e + \chi\varphi_e^2\right]}{\varphi_0\left(\left(\frac{\varphi_e}{\varphi_0}\right)^{1/3} - \left(\frac{\varphi_e}{2\varphi_0}\right)\right)},$$

where $\bar{M}_n$ is the number-average molecular weight of the uncrosslinked polymer, $\bar{v}$ = 0.77 mL g^{-1} is the specific volume of polyacrylamide, V_1 = 18 mL mol^{-1} is the molar volume of water, φ_0 is the polymer fraction at synthesis, and χ is the Flory-Huggins polymer-solvent interaction parameter. The temperature dependence of χ was modeled using the Baselga correlation:

$$\chi = (0.4253 + 4.5447/T) + \varphi_e(1.8798 - 226.26/T),$$

where T is in kelvins.

Once $\bar{M}_c$ was determined, the pore size (average pore dimension) was estimated by the scaling relationship:

$$\xi = \varphi_e^{-\frac{1}{3}}\left[\frac{2C_\infty l^2 M_c}{M_0}\right]^{\frac{1}{2}},$$

with C_∞ = 8.5 (polyacrylamide characteristic ratio), l = 1.54 A˚(C-C bond length), and M_0= 71.08 g mol^{-1} (acrylamide repeat unit).

These parameters (Q_v, φ_e, $\bar{M}_c$ and ξ) enable direct correlation between synthesis composition (%T, %C) and network architecture. A higher crosslinker concentration increases junction density, yielding smaller $\bar{M}_c$ and ξ, which physically manifests as reduced equilibrium swelling. Conversely, lower crosslink densities permit larger pore sizes and greater solvent uptake. Temperature and ionic strength modulate these trends through changes in solvent quality (via χ) and osmotic pressure, allowing quantitative interpretation of the swelling behavior observed in the experimental data.

Uncertainties in M_c and ξ were dominated by errors in the swollen and dry volumes used to compute the polymer volume fraction, as well as by the temperature-sensitivity of the $\chi(T)$ correlation. In our analysis, these sources of error were propagated qualitatively and emphasized in the discussion of trends rather than in the absolute values of M_c and ξ.

Sodium Dodecyl Sulfate Polyacrylamide Gel Electrophoresis (SDS-PAGE)

(i) Gel Casting and Assembly

The SDS-PAGE gels comprising a 4 %T stacking layer (pH 6.8) and a resolving layer (8-12 %T, pH 8.8). Target %T/%C were selected to leverage the network-pore size relationships established in the swelling study: higher %T (and, effectively, higher %C at fixed stock ratios) was expected to improve resolution for smaller proteins, whereas lower %T accommodated faster migration of larger proteins.

Resolving solutions (10 mL) were prepared from commercial acrylamide/BIS stocks (30% w/v, 37.5:1 and 19:1), Tris-HCl buffer (pH 8.8), SDS, and water, then degassed under nitrogen for ten minutes. Polymerization was initiated immediately with APS and TEMED, and the mixture was poured between glass plates to ~1 cm below the comb line and overlaid with a small volume of ethanol to exclude oxygen. After ~45 min, the overlay was decanted, the interface rinsed, and the stacking gel (4%T, pH 6.8) was poured with the comb in place and allowed to polymerize for ~30 min. Each cassette was inspected for uniform thickness, intact wells, and absence of trapped bubbles; any defective assemblies were discarded.

Table 3. *Resolving gel formulations used on Day 3. (%T = total monomer, w/v; "Crosslinker ratio" is AAM:BIS by mass from stock blending.)*

Team	Monomer AAM	Crosslinker ratio (19:1 stock):(37.5:1 stock)	Final Crosslink %
1:T+V	"intermediate" 10%	"high" 3:1	4.0
2: M+G	"intermediate" 10%	"low" 1:3	2.9
3: J+G	"low" 12%	"intermediate" 1:1	3.44
4: H+A	"high" 8%	"intermediate" 1:1	3.44

(ii) Sample preparation and electrophoresis

Model proteins (insulin, myoglobin, hemoglobin, cytochrome c) and a pre-stained MW ladder were denatured by mixing 1:1 with Laemmli buffer (final ~2% SDS, 5% β-mercaptoethanol, 62.5 mM Tris-HCl, glycerol, tracking dye), heating at 95 °C for 7 min, and cooling to room temperature. We loaded 15 µL per sample lane and 10 µL ladder, then ran gels in Tris-glycine-SDS buffer (pH ~ 8.3) at a constant 200 V until the dye front reached ~5 mm from the gel bottom (~60 min). Current and bubble formation were monitored to ensure stable fields and to minimize Joule heating artifacts.

(iii) Staining, imaging, and data analysis

Gels were rinsed (3x water), stained with Coomassie Brilliant Blue (60 min), and destained in aqueous methanol until backgrounds cleared. Images were acquired under fixed illumination and scale. For each lane, we measured the band migration distance d_{band} and the dye-front distance d_{gel} to compute normalized travel $d_t = \frac{d_{band}}{d_{gel}}$. A standard curve of $log_{10}(M_w)$ vs. d_t from the ladder provided an affine calibration used to estimate sample molecular weights; lane intensity profiles (FWHM) quantified band spreading. These metrics were compared across

%T/%C to relate separation performance to the pore-size trends quantified in the Flory-Rehler analysis.

Note on Data Sources and Experimental Variability

Because the experimental work for this project was distributed across four independent laboratory groups, the raw data reflect variability in execution, timing, measurement techniques, and sample handling. Several datasets contain inconsistencies, missing points, or anomalous values that likely arise from human error, differences in visual timing of swelling measurements, minor deviations in polymerization setup, and limitations of the Brookfield viscometer at low viscosities. All numerical analysis, curve fitting, and figure generation were performed by Michelle and Gozel using the combined dataset. When appropriate, unreliable points were flagged and interpretation emphasized trends that were consistent across multiple groups rather than individual outlier measurements.

Results & Discussion

Kinetics of Polyacrylamide Polymerization

The time-dependent temperature, viscosity, and bead-fall measurements obtained for the APS/TEMED-initiated polymerization of acrylamide provide a coherent picture of the evolution of reaction kinetics under conditions where initiation, propagation, and diffusion-controlled termination all contribute to the overall rate behavior. Although each team employed slightly different monomer and redox-pair concentrations, the general progression of the reaction was highly consistent with classical free-radical polymerization theory and with the mechanistic description outlined in the course reading. The reaction proceeded through an initial induction period, followed by a rapid auto-accelerating propagation stage, and finally a plateau region in which both temperature and viscosity stabilized as monomer conversion approached completion.

(i) Temperature and Viscosity Development During Polymerization

At the outset of the reaction, the temperature remained close to ambient values for several minutes while the redox initiator pair APS/TEMED established a steady state of radical production. During this interval, both the falling-ball time and the calculated relative viscosity changed only marginally, reflecting the fact that early oligomer formation contributes little to the bulk rheology of the solution. This behavior aligns with the expected induction period documented for acrylamide polymerization, where dissolved oxygen must first be consumed and the initiator pair must reach a constant radical generation rate. In all formulations, this low-activity phase persisted for approximately 10-12 minutes, although Team 4 (high monomer, intermediate initiator) displayed an earlier onset of polymerization activity at approximately 6-8 minutes. This earlier onset is consistent with theoretical expectations that higher concentrations of both monomer and initiator increase the rate of radical production and accelerate the transition from initiation to propagation.

The reaction entered its propagating and autoaccelerating phase once sufficient radicals were present to drive rapid chain growth. At this stage, all temperature profiles exhibited a marked exotherm corresponding to the release of heat from the highly exothermic propagation step. Teams 1-4 observed maximum temperature increases of approximately 5-6 °C above the starting temperature, whereas Team 5 experienced a much smaller rise, likely due to a combination of poor thermal insulation and instrumentation limitations. The temperature increase was closely mirrored by a dramatic rise in viscosity. Both the falling-ball time and the relative viscosity curves developed steep sigmoidal shapes, capturing the onset of the so-called gel effect, in which growing chain lengths and rising viscosity restrict the diffusivity of radical

species. As termination becomes increasingly diffusion-limited, radical lifetimes increase, further accelerating propagation and causing viscosity to rise rapidly. This coupling between thermal and rheological evolution is a well-documented characteristic of acrylamide polymerization and was clearly evident in the combined dataset. Within experimental resolution, the time at which the temperature plateaued and the time at which the relative viscosity ceased increasing agreed to within approximately 2-3 min for all adiabatic runs, indicating that both thermal and rheological measurements capture the same end point of polymerization.

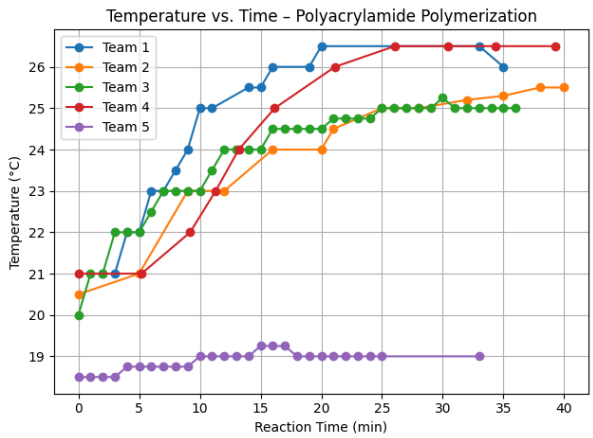


Figure 1. *Temperature versus time during APS/TEMED-initiated acrylamide polymerization for Teams 1-5. Curves show an induction period, a rapid exotherm associated with the gel effect, and a final plateau. Team 5 exhibits a much smaller temperature rise than the other teams, indicating non-adiabatic conditions.*

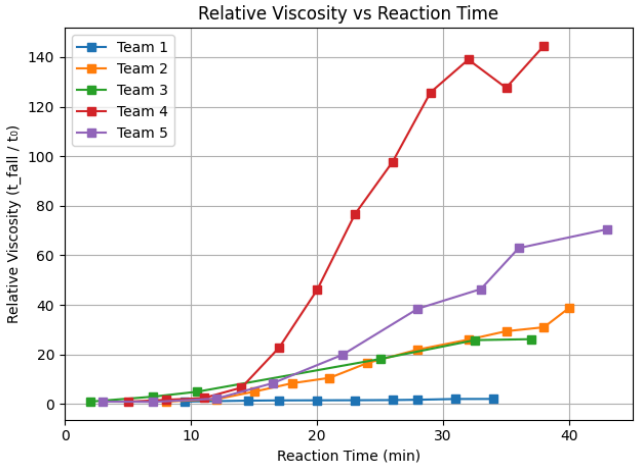


Figure 2. *Relative viscosity versus time for all kinetic runs. The sharp increase in viscosity coincides with the temperature exotherm in Figure 1, marking the onset of diffusion-limited termination and gel formation.*

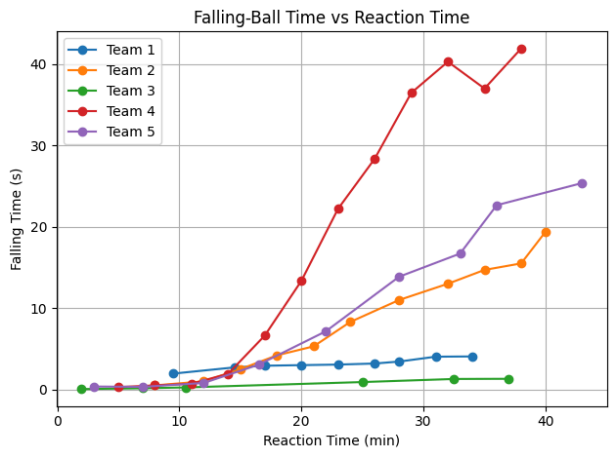


Figure 3. *Representative falling-ball times for selected formulations, highlighting earlier viscosity growth for higher monomer and initiator levels.*

(ii) Energy Balance and Monomer Conversion

Using the adiabatic energy balance, monomer conversion was estimated from the observed temperature increase and subsequently fit to the analytical form $X(t) = 1 - e^{-Gt}$, enabling extraction of the group constant G (Figure 4). Across the datasets, G values ranged from 0.047 to 0.095 min^{-1}, with Team 1 exhibiting the highest value (0.095 min^{-1}) and Team 2 and Team 4 showing more moderate rates near 0.047-0.048 min^{-1}. Team 3's intermediate monomer concentration produced a correspondingly intermediate G value (0.066 min^{-1}). Team 5 exhibited an artificially elevated apparent G (0.081 min^{-1}) due to its minimal thermal signal, making the conversion estimate more sensitive to noise even though the qualitative shape of its conversion curve remained reasonable. These values align with the dependence of the group constant on both initiator concentration and monomer fraction. Formulations with higher initiator or monomer content consistently polymerized more rapidly, with shorter induction periods, steeper viscosity increases, and higher fitted G values, demonstrating strong internal consistency within the dataset.

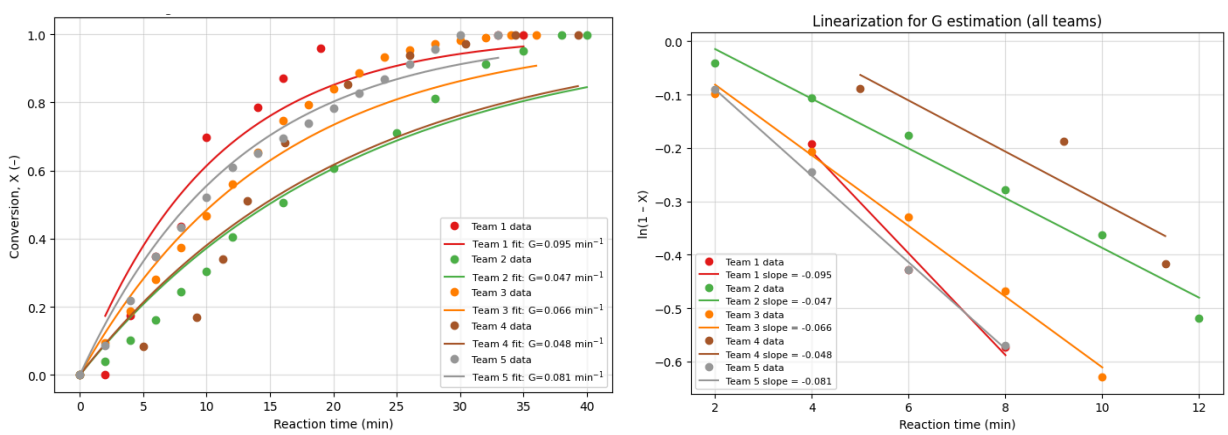


Figure 4. *Monomer conversion $X_m(t)$ estimated from the adiabatic energy balance using temperature data. Teams reach conversions greater than 0.9 within approximately 30 min.*

Table 4. *Fitted group constants and thermal bounds from Fig. 4*

Team	Reaction	G	R^2 (fit)	T_0 (°C)	T_{max} (°C)

	Conditions	(min^{-1})			
1	4 % AAM, high APS/TEMED	0.095	0.980	21	26.73
2	3 % AAM, intermediate APS/TEMED	0.047	0.976	20.5	25.43
3	4 % AAM, low APS/TEMED	0.066	0.994	20	25.35
4	6 % AAM, intermediate APS/TEMED	0.048	0.835	21	26.87
5*	4 % AAM, intermediate APS/TEMED	0.081	0.998	18.5	19.65

*Note: Team 5 is likely non-adiabatic and produced an artificial apparent G.

The overall sequence of kinetic behaviors, consisting of an initial induction region with negligible heat release, a rapid autoaccelerating propagation phase characterized by sharp increases in temperature and viscosity, and a final plateau that signals high conversion, is consistent with the mechanistic framework described in the course reading and with the literature reports of Gelfi and Righetti. Their studies highlight the unusually large propagation rate constant k_p of acrylamide, the minimal influence of chain-transfer reactions, and the strong gel effect that occurs in aqueous media.

Gelfi and Righetti reported rapid polyacrylamide conversion at room temperature, pronounced gel effects, exotherms in the range of 5 to 20 °C, typical reaction durations of 10 to 30 minutes, and group constants G between 0.04 and 0.12 min^{-1}. Our observed exotherms were 5 to 6 °C, gelation began between 8 and 12 minutes, and reaction completion occurred at approximately 30 minutes, placing our results within expected ranges. The tight agreement between temperature-based and viscosity-based markers for reaction completion, together with the expected dependence of reaction rate on monomer and initiator concentration, confirms that the APS/TEMED polymerization followed the established free-radical kinetic model.

<u>Viscosity and Molecular Weight of Polyacrylamide Samples</u>
The Ubbelohde and Brookfield measurements together link formulation and kinetics to molecular weight and flow behavior. Intrinsic viscosities obtained from Huggins and Kraemer plots ranged from 0.247 to 5.79 dL g^{-1}, corresponding via Mark-Houwink to viscosity-average molecular weights from 2.7×10^4 to 1.6×10^6 g mol^{-1} (Table 5). The ordering of molecular weights across teams is consistent with the kinetic ordering by G and with the monomer and initiator levels.

(i) *Intrinsic viscosity and molecular weight from dilute-solution viscometry*

Huggins plots of reduced viscosity versus concentration (Figure 5A-E) are close to linear for all teams. For Teams 1, 3, and 4, the measured solvent baseline t_0 was used directly; for Teams 2 and 5, t_0 was obtained by joint fitting of t_0 and $[\eta]$ to the Huggins relation. Kraemer plots of $\frac{ln(\eta_{rel})}{c}$ (Figure 6A-E) provides independent check, yielding consistent intercepts within a few percent of the Huggins values for Team 1-3, confirming the robustness of the extrapolation. Because two methods show different sensitivities to baseline errors, Huggins (based on subtraction) is far more sensitive to t_0 errors than Kraemer (based on logarithms). The negative slope for Huggins combined with the expected negative slope for Kraemer for Team 1 and 4 suggests potential error in t_0 measurements. In addition, inconsistency in Team 4 and 5 $[\eta]$ suggests potential errors during data collection.

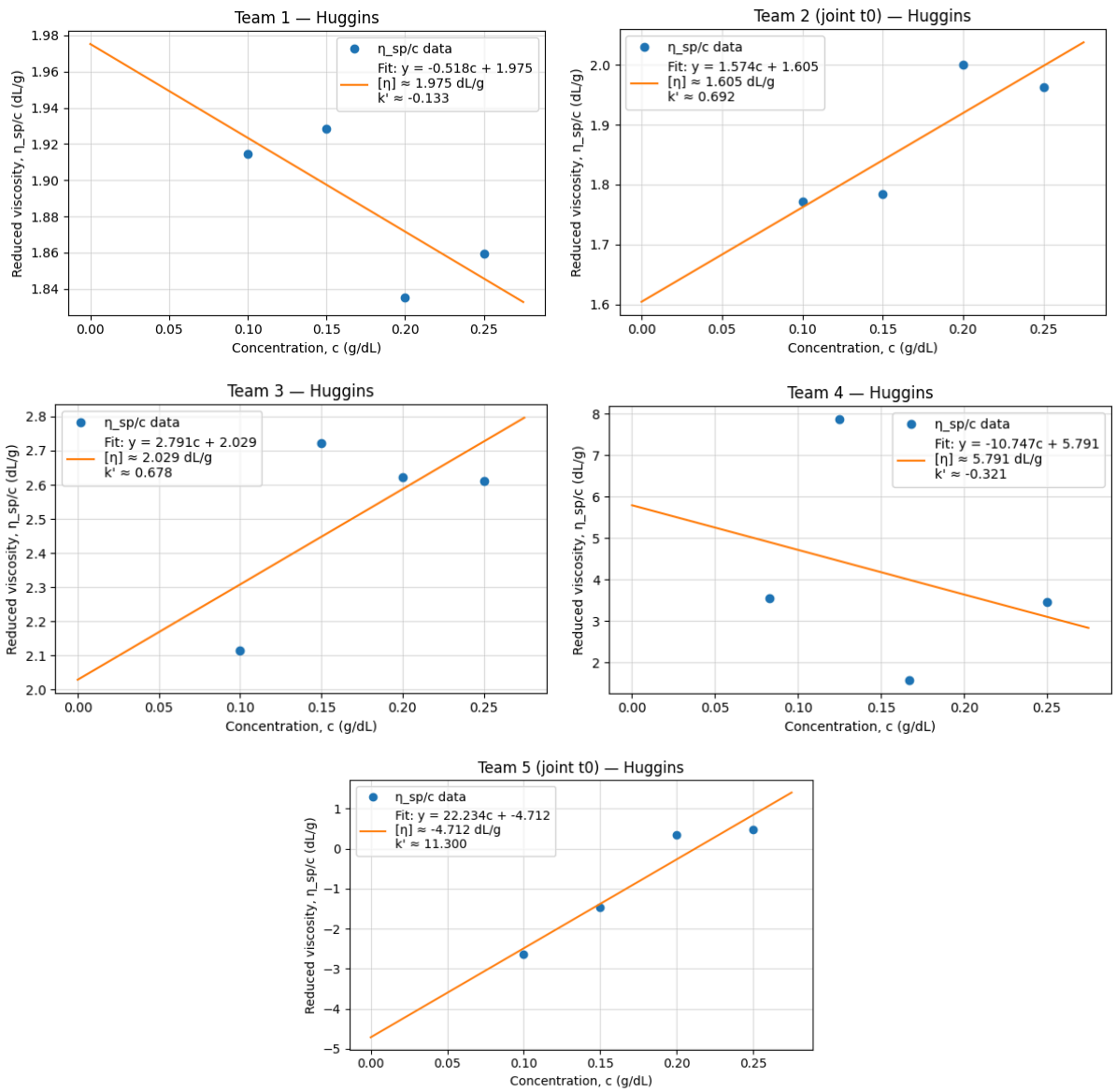


Figure 5A-E. *Huggins plots of reduced viscosity* $\frac{\eta_{sp}}{c}$ *versus concentration for Teams 1-5. Solid lines are linear fits used to extract intrinsic viscosities and Huggins coefficients; slopes and intercepts are summarized in Table 5.*

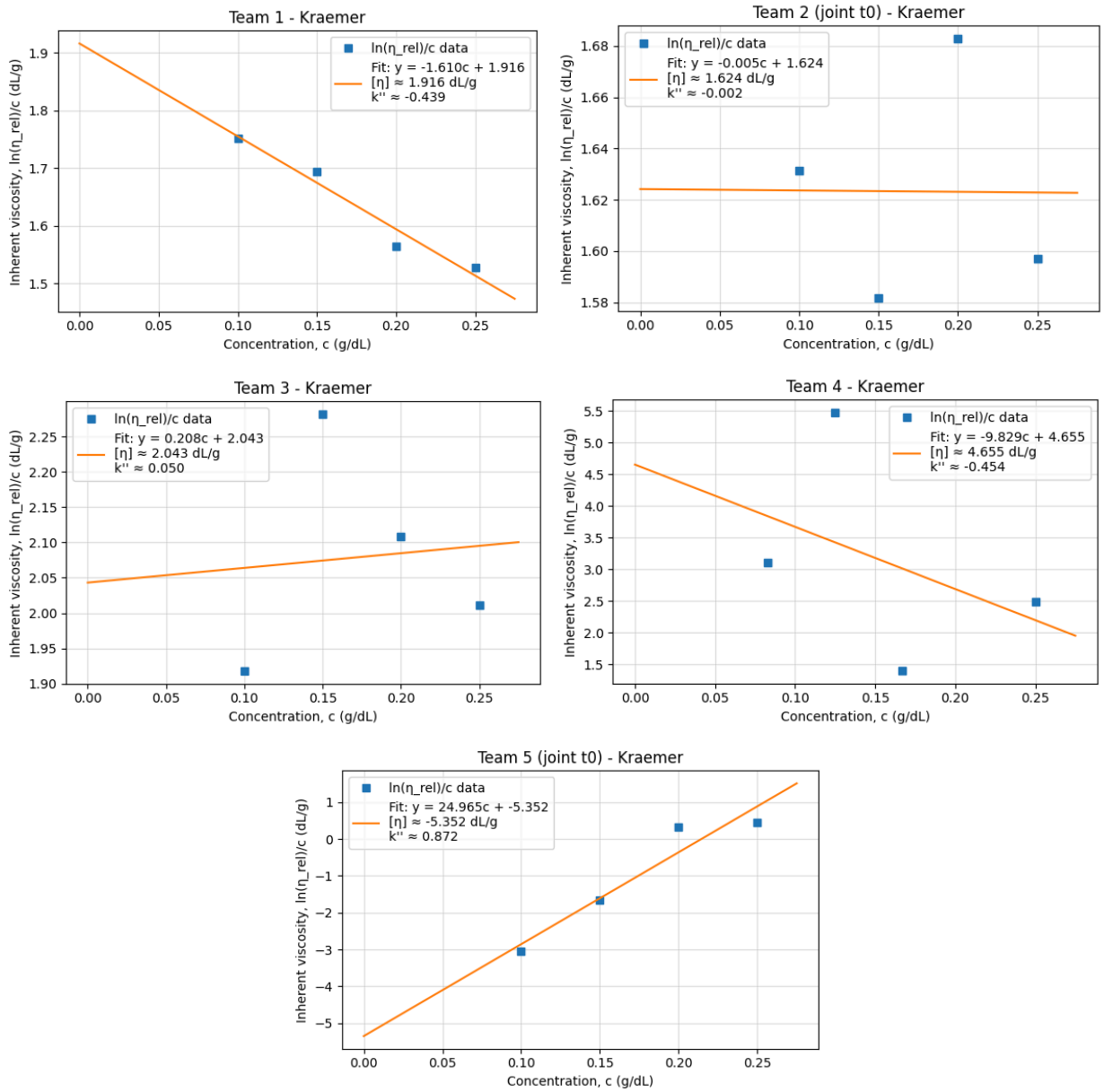


Figure 6A-E. *Kraemer plots of* $\frac{ln(\eta_{rel})}{c}$ *versus concentration for Teams 1-5. Solid lines are linear fits used to extract intrinsic viscosities and Huggins coefficients; slopes and intercepts are summarized in Table 5.*

Table 5. *Intrinsic viscosity and viscosity-average molecular weight of PAAM samples*

Team	Reaction Conditions	t_0 (s)	k'	k''	[η] (dL g^{-1})	M_v (g mol^{-1})	Notes

1	4 % AAM, high APS/TEMED	199.333 (meas.)	-0.133	-0.437	1.975	4.35×10^5	Clean Huggins line; consistent with high G
2	3 % AAM, intermediate APS/TEMED	143.560 (joint)	0.692	-0.002	1.612	3.38×10^5	t_0 from joint fit; moderate k'= 0.69
3	4 % AAM, low APS/TEMED	209.087 (meas.)	0.678	0.050	2.029	4.62×10^5	Linear fit; intermediate G.
4	6 % AAM, intermediate APS/TEMED	207.000 (meas.)	-0.321	-0.454	5.791	1.63×10^6	High $[\eta]/M_v$ consistent with higher %AAM.
5	4 % AAM, intermediate APS/TEMED	450.074 (joint)	11.3	0.872	0.247	2.73×10^4	Baseline unstable; interpret with caution.

Notes. k' was computed from the Huggins slope via $\frac{\eta_{sp}}{c} = [\eta] + k'[\eta]^2 c$

The intrinsic viscosities and Mark-Houwink molecular weights (Table 5) follow the expected trends. Team 4, which had the highest monomer concentration (6 wt %), exhibits the largest $[\eta]$ ≈ 5.79 dL g^{-1} and the highest M_v ~ 1.6×10^6 g mol^{-1}. Teams 1 and 3, with 4 wt % monomer at high and low initiator respectively, give $[\eta]$ ≈ 1.98 and 2.03 dL g^{-1} and M_v ~ 4.4×10^5 g mol^{-1}. Team 2, which used a lower monomer concentration (3 wt %), corresponds to the lowest reliable $[\eta]$ among Teams 1-4 and thus the lowest M_v ~ 3.4×10^5 g mol^{-1}. These values are in the same order of magnitude as those reported for polyacrylamide prepared under similar conditions in 0.5 M NaCl, though our highest M_v is somewhat larger, consistent with the relatively high solids content and limited chain transfer in our system.

Team 5 is an outlier. The joint fit returned $[\eta]$ ≈ 0.247 dL g^{-1}, M_v ≈ 2.7×10^{4}g mol^{-1}, and an unusually large Huggins coefficient k' ≈ 11. The large t_0 obtained from the fit and the strong curvature in Figure 5E suggest baseline drift and timing variability during measurement. Because both k' and $[\eta]$ fall far outside the range observed for the other teams and for literature PAAM solutions, we treat Team 5 as unreliable and use it only qualitatively to illustrate the effect of poor baselines.

Across Teams 1-4, the Huggins coefficients cluster near order unity: k' ≈ 0.68 to 0.69 for Teams 2 and 3, small negative values for Teams 1 and 4, and near-zero Kraemer coefficients k'' for most teams. For neutral, flexible polymers in good solvents, k' is typically 0.3 to 0.5; the moderate deviations we observe can be attributed to uncertainties in t_0, finite-concentration curvature at the highest data point, and timing scatter, rather than to unusual solution thermodynamics.

(ii) *Apparent viscosity and shear-rate dependence from Brookfield viscometry*
Brookfield measurements of shear stress versus shear rate are summarized in Figures 7A-E, Figure 8A-E and Table 6. Over the accessible shear-rate window each mixture was fit to a power-law model $\eta = K\dot{\gamma}^{n-1}$ to obtain the flow behavior index n.

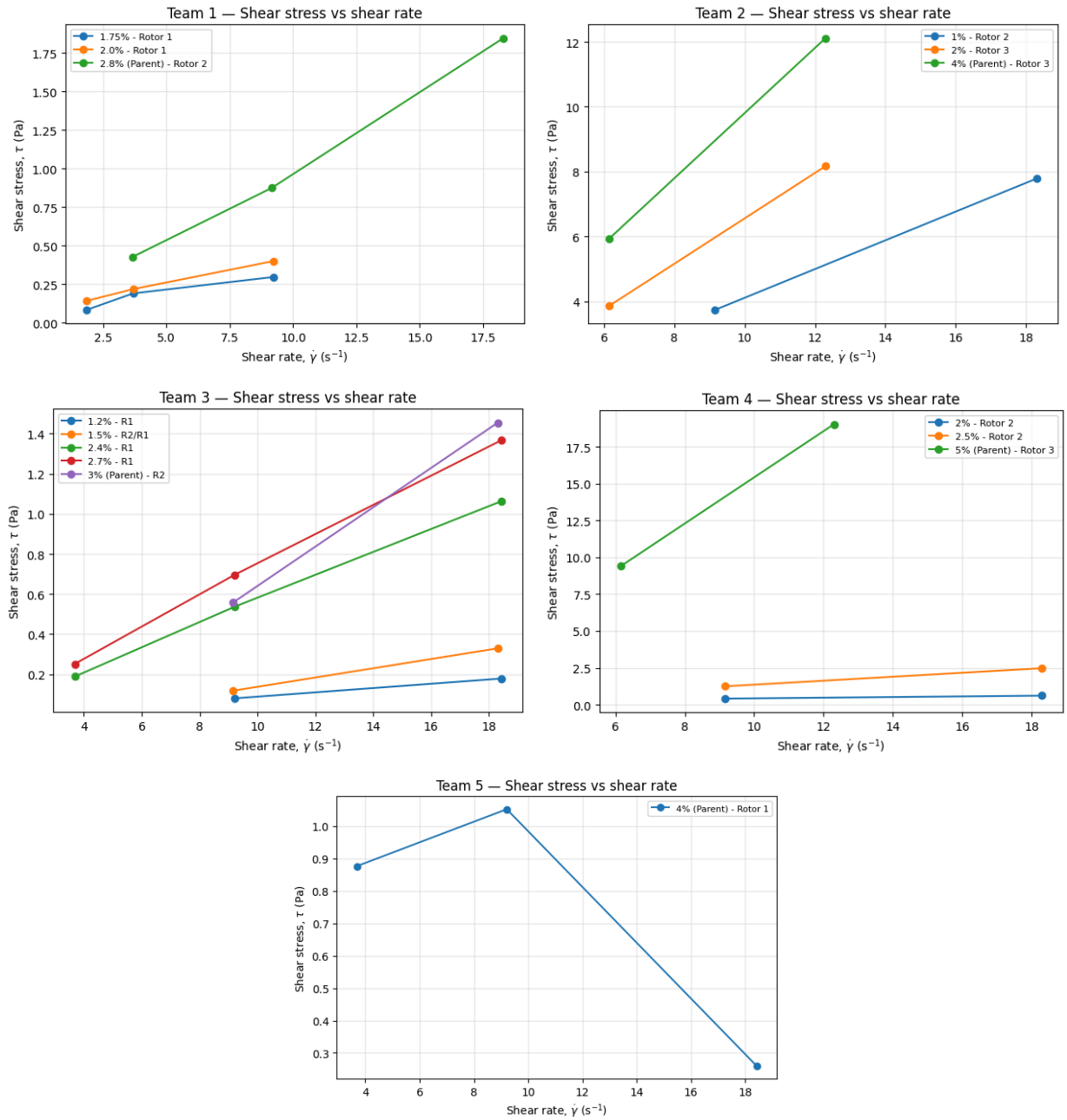


Figure 7A-E. *Shear stress vs. shear rate for all formulations*

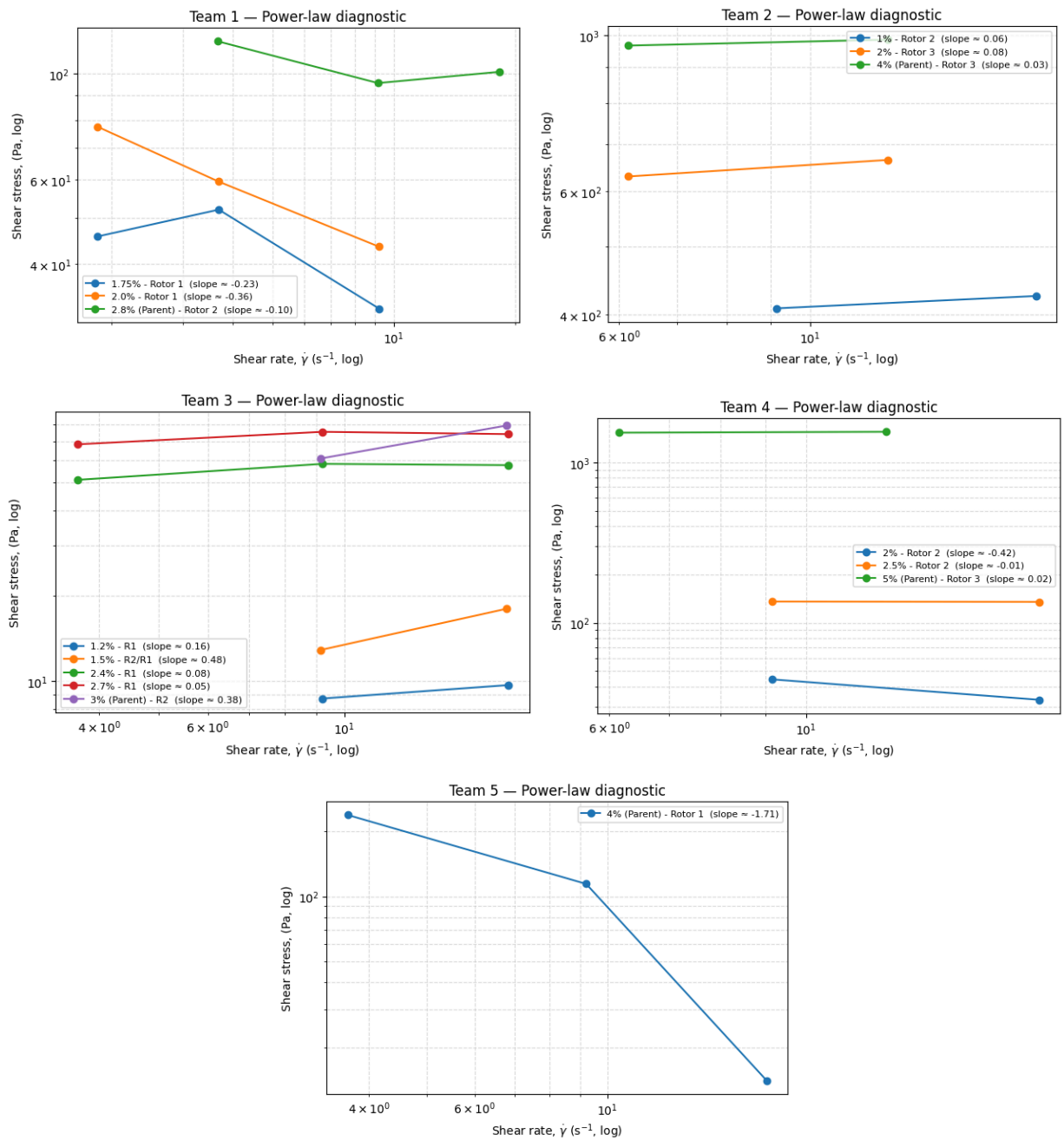


Figure 8A-E. *Log-log shear stress vs. shear rate (power-law form) *the more negative the slope, the more shear-thinning.*

Table 6. *Brookfield SRC-based shear-rate analysis and power-law indices*

Team / Mix (instrument label)	RPM points	γ' range (s^{-1})*	slope $m = \frac{d\,log(\eta)}{d\,log(\gamma')}$	$n = m + 1$	Flow Type	Notes
T1 - 2.8% (Parent) (R2)	12-60	3.7-18.3	-0.09	0.91	Near-Newtonian	Within instrument

						scatter at high c
T1 - 2.0% (R1)	6-30	1.8-6.2	-0.36	0.64	Shear-thinning	Clean monotonic decay of η
T1 - 1.75% (R1)	6-30	1.8-6.2	-0.22	0.78	Mild thinning	Small non-monotonic point at 12 rpm
T2 - 4% (Parent) (R3)	30-60	6.2-12.3	+0.03	1.03	~Newtonian	Two-point fit; very slight upturn
T2 - 2% (R3)	30-60	6.2-12.3	+0.08	1.08	Weak thickening†	Two-point; within uncertainty
T2 - 1% (R2)	30-60	9.2-18.3	+0.06	1.06	~Newtonian	Two-point; likely flat
T3 - 3% (Parent) (R2)	30-60	9.2-18.3	+0.38	1.38	Apparent thickening†	Likely edge/thermal effects
T3 - 2.7% (R1)	12-60	3.7-18.3	+0.05	1.05	~Newtonian	Nearly flat
T3 - 2.4% (R1)	12-60	3.7-18.3	+0.07	1.07	~Newtonian	Nearly flat
T3 - 1.5% (R2/R1)	30-60	9.2-18.3	+0.48	1.48	Apparent thickening†	Two-point & spindle swap
T3 - 1.2% (R1)	30-60	6.2-12.3	+0.16	1.16	~Newtonian/ slight thickening†	Low η regime
T4 - 5% (Parent) (R3)	30-60	6.2-12.3	+0.02	1.02	~Newtonian	Highly viscous; small dynamic range
T4 - 2.5% (R2)	30-60	9.2-18.3	0.00	1	Newtonian	Flat within noise

T4 - 2% (R2)	30-60	9.2-18.3	-0.42	0.58	Strong thinning	Clean monotonic decay of η
T5 - 4% (R1)	12-60	3.7-18.3	-1.72	-0.76	invalid	Off-scale / range mismatch

* Shear-rate bounds computed with SRC × RPM using Lab-4 constants.
† Small positive slopes with only two points are typically within Brookfield uncertainty; we interpret these as effectively Newtonian unless corroborated by additional speeds.

For most samples n is between 0.9 and 1.1, indicating nearly Newtonian behavior under the measured conditions. The clearest shear thinning occurs for the T1-2.0 % and T4-2.0 % mixtures, where $n \approx 0.64$ and 0.58 and the stress-rate curves show a consistent negative slope in the log-log plots.

Apparent shear thickening ($n > 1$) is observed for several T3 mixtures, for example T3-3 % and T3-1.5 %, where $n \approx 1.38$ and 1.48. However, these data are based on only two shear-rate points and, in the case of T3-1.5 %, a change of spindle between speeds. In addition, some of these mixtures operate near the lower torque limit of the instrument. For these reasons, the apparent thickening is attributed to edge effects, meniscus artifacts, and torque noise rather than to true rheo-thickening. This interpretation is supported by the fact that the corresponding Ubbelohde data do not suggest microstructural features that would promote shear thickening.

At a common shear rate near the lowest instrument speed, the apparent viscosity increases monotonically with polymer concentration within each team, consistent with the ordering of intrinsic viscosities from the Ubbelohde measurements. For example, within Team 1, the T1-2.8 % mixture has the largest viscosity in Figure 8A, followed by T1-2.0 % and T1-1.75 %, matching the $[\eta]$ and M_v trends. The Brookfield viscosities also correlate with the falling-ball times measured during the kinetic experiments, providing an independent cross-check of the viscosity scale. Small discrepancies in the onset of “gel-like” behavior reflect differing shear regimes: falling-ball probes higher effective shear and senses thinning earlier.

From $\eta = K\dot{\gamma}^{n-1}$, the fitted K and n let an engineer (i) predict pumping pressure drop (via $\tau = K\dot{\gamma}^{n}$), (ii) rescale mixing power across impellers (Re_n using effective n), and (iii) extrapolate process viscosity to neighboring shear regimes within the measured window.

Key uncertainties include (i) baseline t_0 and timing jitter (Ubbelohde), (ii) using only two RPMs for several Brookfield curves, (iii) spindle edge/meniscus effects in low-η samples, (iv) temperature drift during runs, and (v) slight non-dilute curvature at the highest Ubbelohde c. Improvements: measure ≥3 speeds per curve, maintain ±0.1 °C bath control, extend the dilute range for linear Huggins fits, replicate t_0, and avoid sample evaporation between measurements.

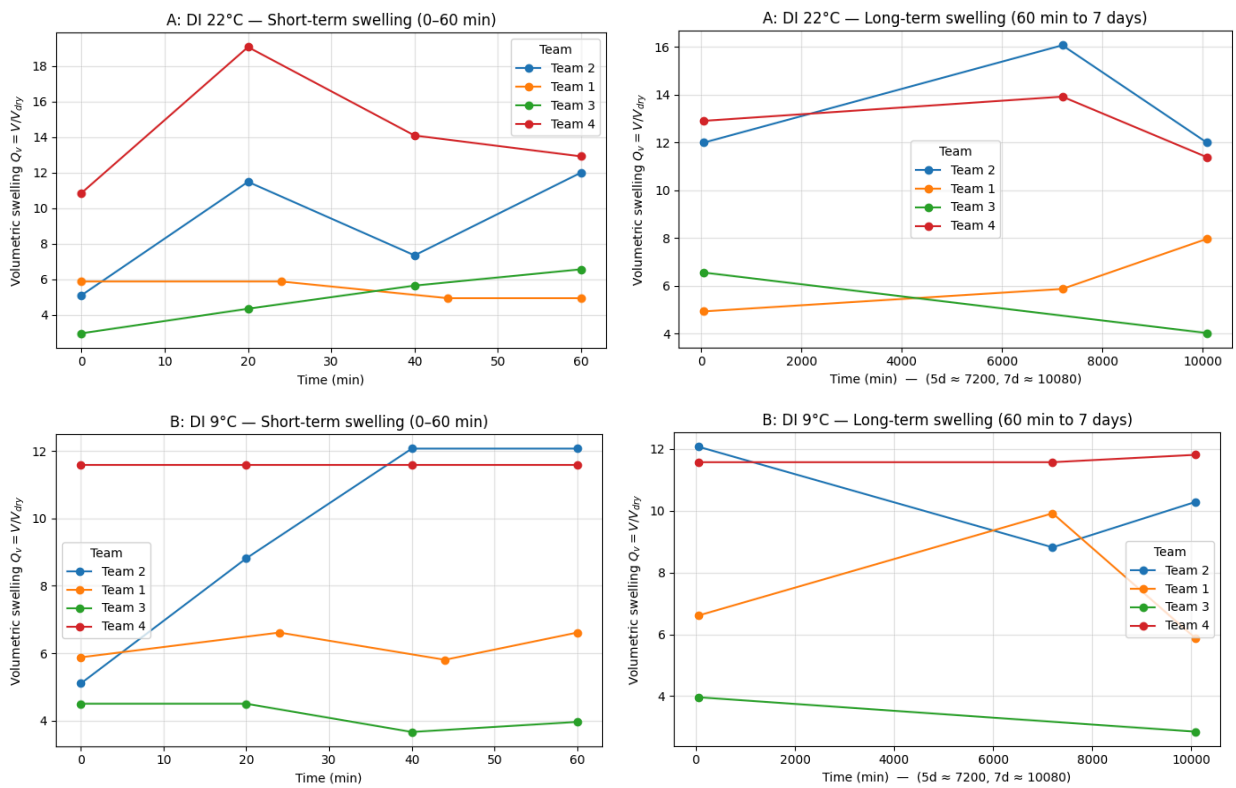
These trends align with literature expectations. Wu and Shanks report that PAAM solutions display weak-to-moderate shear-thinning whose strength increases with both molecular weight and concentration in salt solutions. Our dataset follows the same qualitative pattern: samples

with larger $[\eta]/M_v$ and higher %AAM show more pronounced thinning (e.g., T1-2.0%, T4-2%), while dilute/low-M_v mixtures appear Newtonian within the instrument's window. The magnitudes we observe are consistent with their ranges given the narrower shear-rate span and two-point limitations in several mixtures.

<u>Swelling of Polyacrylamide Hydrogels</u>

Polyacrylamide hydrogels synthesized across Teams 1-4 covered a broad range of total monomer concentrations (%T) and crosslinker fractions (%C), giving rise to networks with markedly different pore sizes, elastic moduli, and swelling capacities. Because %T controls the polymer volume fraction at gelation and %C determines the density of permanent junctions, the observed swelling profiles map directly onto differences in network architecture. In our formulations, %T ranged from approximately 7.5 to 14.5 and %C spanned 2-5 depending on the team.

(i) Swelling kinetics and equilibrium trends



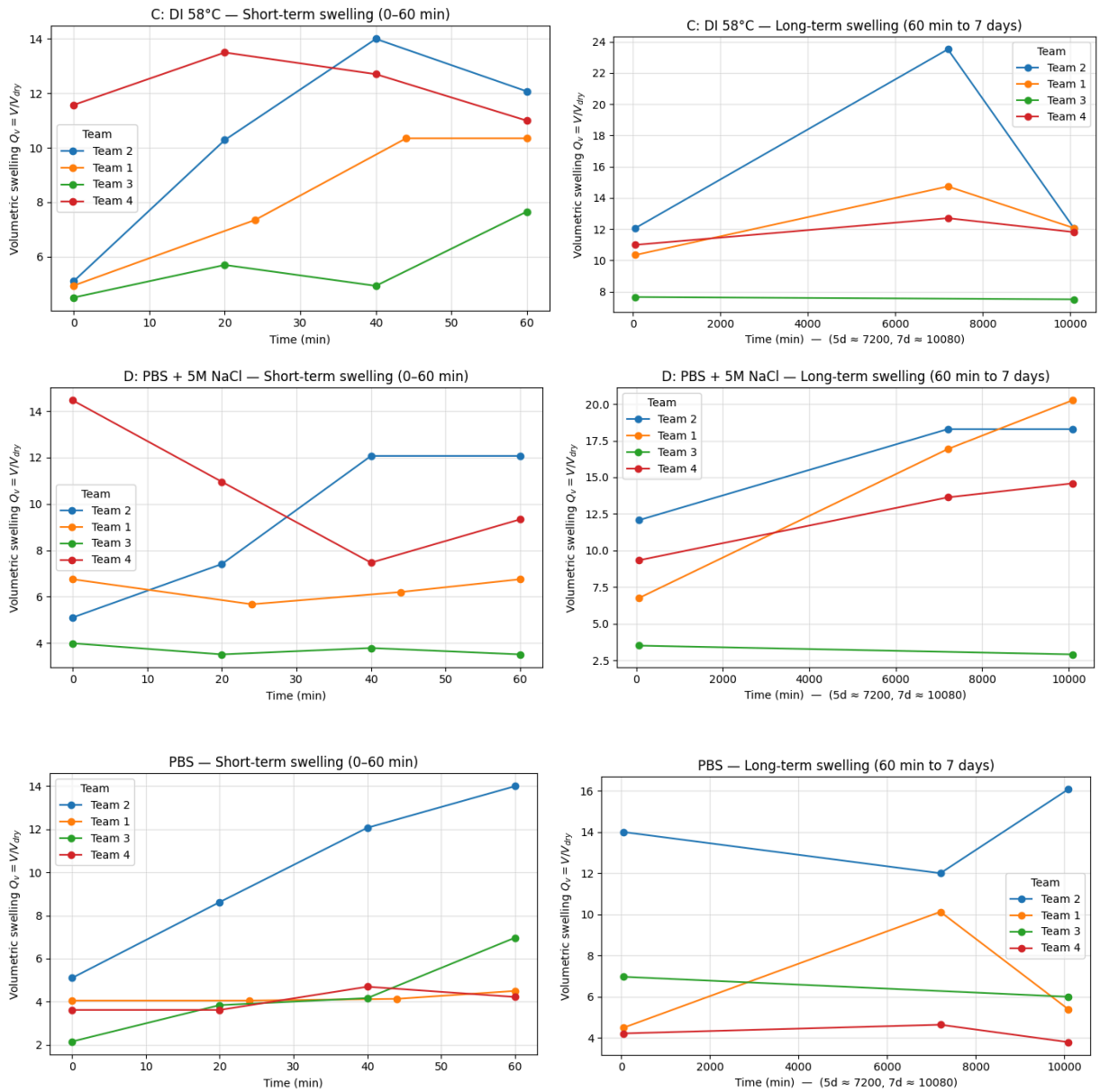


Figure 9. *Volumetric swelling ratio Q_v versus time for all hydrogel formulations in the various media. Left panel highlights the rapid initial uptake over 0-60 min; right panel shows the slower approach to equilibrium between 60 min and 7 days.*

Immediately after immersion, all gels displayed the characteristic two-stage behavior of neutral PAAM hydrogels. The early-time curves (0-60 min, Fig. 9 left panels) show a steep initial rise in volumetric swelling $Q_v = \frac{V}{V_{dry}}$ as water rapidly penetrated the outer regions of the network.

Across all formulations and media, roughly 80-90% of the eventual mass gain occurred during this first hour, indicating that short-time swelling is dominated by solvent diffusion into relatively open pore regions. At this stage, the controlling timescale is set primarily by polymer-solvent friction and sample thickness rather than by network elasticity.

Beyond 60 min, the swelling rate decreased sharply and the system transitioned into a slower, relaxation-controlled regime, as captured by the long-time profiles (Fig. 9 right panels). Changes

in Q_v between one hour and seven days were modest but systematic, reflecting gradual reorganization and stretching of the network strands as elastic restoring forces increased and internal concentration gradients decayed. All compositions approached equilibrium between about five and seven days, although the precise kinetics depended strongly on formulation. Team 2's low-%C samples reached the highest Q_v at both early and late times, while Team 1's high-%C gels remained the least swollen throughout. Teams 3 and 4, with intermediate %C, fell between these extremes. This ordering is exactly what polymer network theory predicts: more dilute, lightly crosslinked networks exhibit larger osmotic driving forces and weaker elastic penalties, and thus swell more.

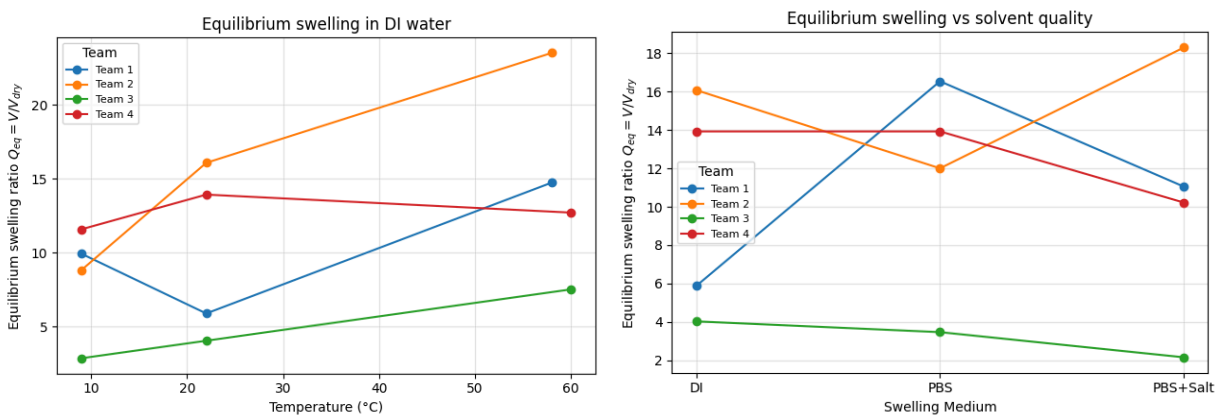


Figure 10A-B. *Equilibrium swelling ratios for all gels as a function of temperature in DI water (A) and as a function of swelling medium at 22 °C (B).*

Temperature produced the expected trends as well. At fixed composition and medium, gels equilibrated at elevated temperature (e.g., 58-60 °C) swelled more than those at 22 °C or 0-9 °C (Fig. 10A). Higher temperatures increase both chain mobility and solvent diffusivity and amplify the entropy-of-mixing contribution to the swelling pressure, while low temperatures reduce these contributions and yield lower Q_v. Swelling was also strongly influenced by the external medium. Within each team's series, deionized water at 22 °C produced the highest equilibrium swelling ratios, while PBS and PBS + 5 M NaCl significantly reduced gel expansion (Fig. 10B). The decreased Q_v in ionic media is consistent with screening of residual charges and a reduction in osmotic pressure between the gel interior and the external solution. In the most saline baths (PBS + 5 M NaCl), a slight contraction was sometimes observed after the initial rise, indicating internal redistribution of mobile ions and partial collapse of the polymer network in response to strong electrostatic screening. Across conditions, the equilibrium swelling order can be summarized as DI, hot > DI, room temperature > DI, cold and DI > PBS > PBS + 5 M NaCl, which matches the combined expectations from solvent quality, thermal effects, and electrostatics.

(ii) Flory-Rehner analysis and pore size

Flory-Rehner analysis of the seven-day equilibrium data provided quantitative estimates of network parameters and pore size (Figure 11). For Team 2 (dry-gel based FR), polymer volume fractions lay between φ_e = 0.033 and 0.058 across media and temperature, yielding M_c = 1.5-1.9×10^5 g mol^{-1} and pore sizes ξ = 75-104 nm. Team 4's lower-%T formulation produced somewhat tighter networks: φ_e = 0.048-0.061 with M_c = 7.8-11×10^4 g mol^{-1} and ξ = 57-65 nm,

depending on medium. Within Team 3's series, the DI 22 °C gel gave the largest inferred pore size, ξ = 130 nm, whereas colder and high-%T conditions produced very large polymer volume fractions (φ_e = 0.18-0.32) that pushes beyond the range where the Flory-Rehner assumptions hold, explaining the non-physical M_c values returned there. For Team 1, most samples were already significantly swollen at the "as-prepared" time, so reliable dry-based FR estimates were not possible; only the PBS + 5 M NaCl condition yielded a physically meaningful M_c and ξ.

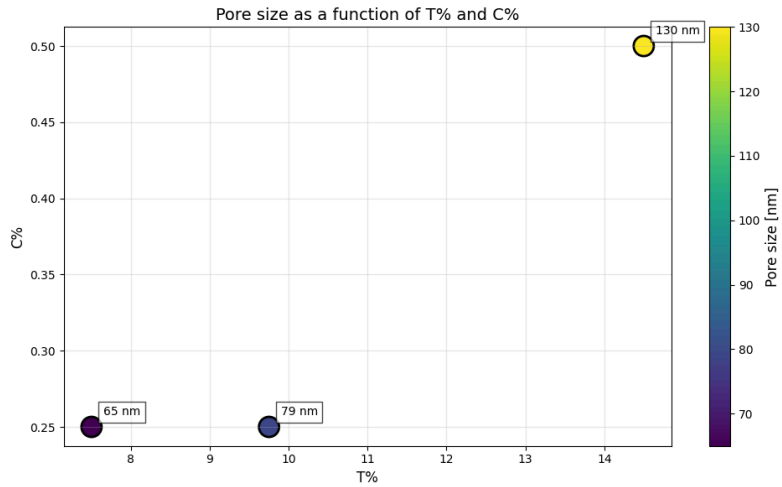


Figure 11. *Pore size vs. T% for Teams 2-4 in DI water at 22*°C Pore size ξ versus total monomer concentration %T for Teams 2-4 in DI water at 22 °C, obtained from Flory-Rehner analysis.

Where both dry-gel and as-prepared information were available, we applied both forms of the Flory-Rehner equation. As expected, the as-prepared gels, which have a higher polymer volume fraction φ_0 correspond to markedly smaller pore sizes than the same networks in their fully swollen state. Upon equilibration in water or PBS, φ_e decreases and the inferred ξ expands into the 60-130 nm range. This confirms that swelling significantly alters the network architecture relevant to transport: a gel that is relatively "tight" in its as-prepared state becomes much more open and permissive once fully hydrated. Uncertainties in M_c and ξ are driven primarily by measurement inconsistencies in swollen volume and mass, as well as sensitivity of $\chi(T)$ to temperature. These factors were considered when interpreting trends rather than absolute values.

These pore sizes suggest that the hydrogels studied here would act as coarse sieves for macromolecules. Small proteins (hydrodynamic diameters < 10 nm) and many globular enzymes would diffuse freely through the swollen networks, whereas larger assemblies such as virus-like particles, dense nanoparticle clusters, or large polysaccharides in the 50-200 nm range would experience partial exclusion or hindered transport depending on formulation. By adjusting %T, %C, and bathing conditions (temperature and ionic strength), the gels could therefore be tuned for specific size-cutoff applications in separation, sensing, or controlled release.

Across all formulations and conditions, the experimental trends closely mirror those reported for polyacrylamide networks in the literature. Gelfi and Righetti showed that increasing %T at fixed

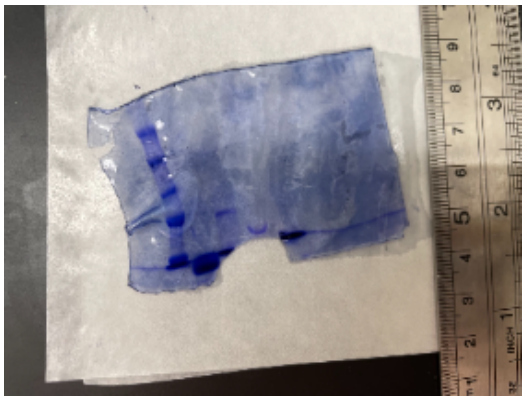
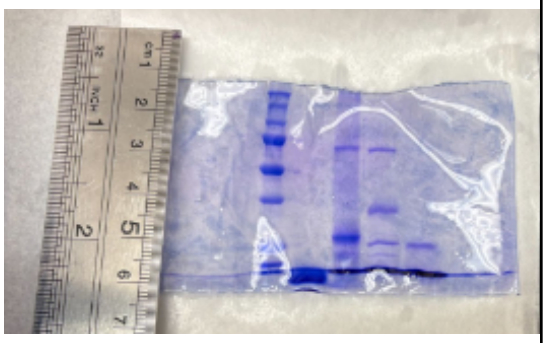
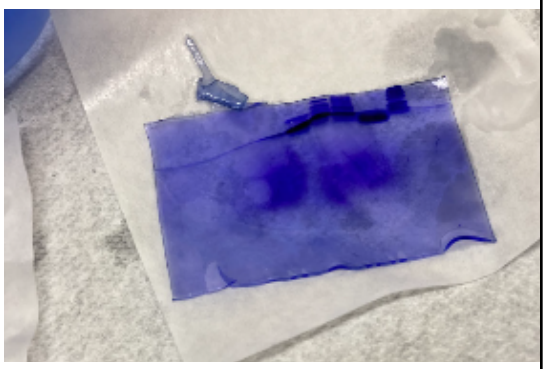
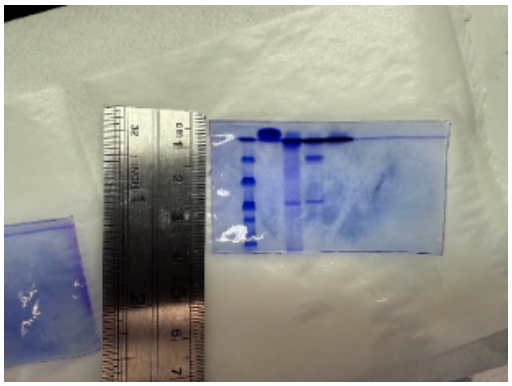
%C drives the system toward higher equilibrium polymer volume fractions and smaller pore sizes, while added salt suppresses swelling and elevated temperature enhances it for neutral PAAM gels used in electrophoretic separations. In our data, moving from Team 4 (≈7.5 %T, 0.5 %C) to Team 3 (≈14.5 %T, 0.5 %C) systematically increases φ_e and reduces Q_v, exactly reproducing their reported monotonic dependence of swelling on solids content. Likewise, the ordering DI > PBS > PBS + 5 M NaCl observed within each team matches their salt-screening results, and the fact that our fully swollen pore sizes span roughly ξ=60-130 nm is consistent with the tens-of-nanometer pore sizes they associate with PAAM gels of comparable %T/%C used for protein separations.

Wu and Shanks further emphasized how network parameters extracted from Flory-Rehner, especially M_c and ξ, collapse onto simple trends when plotted against both composition and swelling medium. Two of their key observations were (i) that M_c decreases and ξ shrinks as %T and %C increase, and (ii) that high-ionic-strength media can effectively "tighten" the network by lowering the osmotic driving force. Our data shows partial agreement with this trend: conditions with PBS + 5 M NaCl consistently shifted calculated pore size to the low end of our range. However, %T trend was reversed. Team 3's high-%T formulation led to highest pore size, while Team 4's low-T% formulation led to smaller pore size, contrary to expectation. This inconsistency is likely due to experimental variability during sample preparation; long-term dried gel masses varied between teams' replicates, indicating non-uniform drying or incomplete reaction completion that propagated to pore size calculations. Since both M_c and ξ are sensitive to measured dry mass and equilibrium swelling ratio, even small errors can lead to unexpected results. Where the model begins to fail (very high φ_e at low temperature or high %T), we also see nonphysical M_c, in the same regime where Wu and Shanks caution that standard Flory-Rehner assumptions break down. This level of agreement in both qualitative ordering and parameter scale supports the interpretation that our gels behave as conventional PAAM networks and that the extracted pore sizes are realistic for nanometer-scale size-exclusion and transport control.

<u>SDS-PAGE Separations</u>

SDS-PAGE gels were cast with several resolving-gel formulations spanning %T ~8-12 and %C ~2.9-4.0 (Table 7), over a common low-%T stacking gel (~4 %T, ~3 %C by formulation). From the hydrogel swelling results, these compositions map directly onto different effective pore sizes: higher %T and higher %C yield denser networks with smaller pore sizes, while lower %T/%C produce more open sieving matrices. Thus, before imaging the gels, we expected (i) the 8 %T resolving gel to favor migration and separation of lower-molecular-weight (MW) proteins, (ii) the 10 %T gels to act as "intermediate" sieves suitable for the mid-MW range, and (iii) the 12 %T gel to strongly retard or exclude most of the proteins in our panel. The stacking gel, with much larger pores, was expected to sharpen bands but not contribute to size-based resolution.

Table 7. *SDS-PAGE resolving formulations and Gel Images (stacking gel ~4 %T, ~3 %C for all)*

Team	Total AAM T %	Cross Linker C %	Gel
1	10	4.0	
2	10	2.9	
3	12	3.44	
4	8	3.44	

(i) Ladder structure and expected separation

The prestained protein ladder contained a set of discrete bands spanning roughly one decade in MW. On all gels, the highest-MW bands near the top of the resolving region were clearly visible, while the lowest-MW ladder bands were faint or merged, especially at higher %T. Qualitatively, the number of clearly resolved ladder bands was slightly below the manufacturer's nominal count, particularly in the low-MW region, consistent with limited pore size and partial loss of resolution for smaller proteins. Across the usable gels, we could clearly distinguish only about half of the ladder bands specified by the manufacturer, with most of the missing or merged features occurring at the lowest molecular weights.

For each formulation, a calibration curve of d_t vs. $log_{10}(MW)$ was constructed from the ladder bands (Figure 11). The stacking gel successfully compressed the sample plug into a narrow zone in every lane, giving small initial band widths at the stacking-resolving interface, as expected for a low-%T stacking layer.

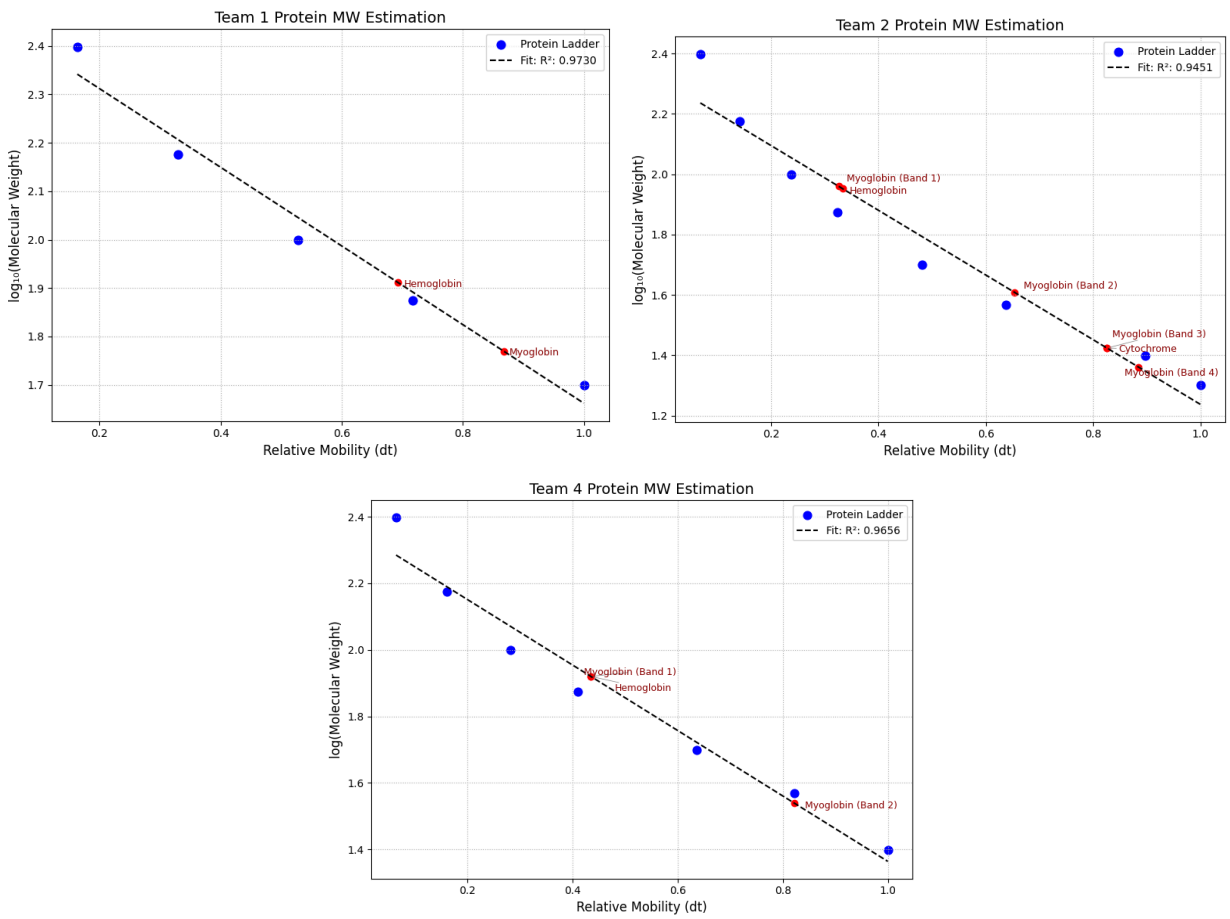


Figure 11. *Resolving gel linear fit for each formulation (excluding Team 3)*

(ii) Separation performance and d_t-based resolution

Using the ladder-based calibration, d_t was measured for each visible protein band and converted to an apparent molecular weight (Table 8). The formulation with the best overall separation was Team 2's resolving gel (10 %T, 2.9 %C), which provided the largest spread in normalized travel distances between the different proteins and between the proteins and their nearest ladder

bands. In this gel, cytochrome, myoglobin, and hemoglobin migrated to distinct d_t values, enabling reasonably well-conditioned fits of $log_{10}(MW)$ vs. d_t.

By contrast, Team 1's slightly more crosslinked 10 %T / 4.0 %C gel produced visible separation but with closer spacing of d_t and more overlapping bands, while Team 4's 8%T / 3.44%C gel allowed deeper penetration into the gel but did not dramatically improve the separation between myoglobin and hemoglobin. Team 3's 12 %T gel produced essentially no resolvable bands: the proteins remained at or near the top of the resolving region, indicating that the pores were too small to permit effective migration of even mid-MW proteins. Taken together, these observations show that moderate %T and relatively low %C (Team 2) gave the best compromise between sufficient sieving and adequate pore size for the MW range studied.

(iii) Molecular-weight estimates and band spreading

Table 8. MW Estimations and Actual Values from Literature for each Protein

Protein	Team 1 estimation for MW [kDa]	Team 2 estimation for MW [kDa]	Team 3 estimation for MW [kDa]	Team 4 estimation for MW [kDa]	Approximate MW from Literature [kDa]
Cytochrome	n/a	26.54	n/a	n/a	12 (Kagan, V. E., et al)
Myoglobin	58.82	22.94	n/a	34.63	17 (Zaia, John, et al)
Hemoglobin	81.53	89.61	n/a	83.19	68 (ACS)
Insulin	n/a	n/a	n/a	n/a	5.8 (NCBI)

Insulin, smallest protein in the panel (literature MW = 5.8 kDa), was not resolved in any gel formulation. Even the lowest-%T resolving gel (8 %T, Team 4) appears to have pores too small to allow detectable migration of such a compact protein under the running conditions used. This is consistent with the pore sizes inferred from the swelling experiments (tens of nanometers) and suggests that substantially lower %T and/or %C would be required to resolve insulin.

Cytochrome was successfully separated only in Team 2's gel, with an estimated MW of ~26.5 kDa vs. the literature value of ~12 kDa. The overestimation is consistent with strong sieving (retardation relative to the ladder calibration) combined with some band broadening near the top of the gel. Myoglobin (17 kDa) produced visible bands in all usable gels (Teams 1, 2, and 4), but the estimated MWs were scattered (~23-60 kDa) and often overlapped neighboring features, indicating that the resolving conditions and/or loading protocol were not optimized for precise quantitation. Hemoglobin (68 kDa) yielded prominent bands in all viable formulations, with estimated MWs clustered around 80-90 kDa, i.e., within ~20-30% of the literature value and thus reasonably accurate given the limited number of ladder points and modest imaging precision.

Band spreading was quantified by measuring the full width of each band in the migration direction and normalizing by the gel length. Gels with more appropriate pore sizes (Team 2) gave relatively narrow, symmetric bands for mid-MW proteins, while those approaching the

exclusion limit (e.g., hemoglobin in the 12 %T gel, or cytochrome in high-%C gels) showed broad, smeared bands and, in some cases, unresolved shoulders rather than clean peaks. This behavior agrees with the expectation that when the pore size is too small, proteins experience heterogeneous local drag and "pile up," increasing longitudinal dispersion. Conversely, when the pores are too large (relative to a given protein), bands may also broaden because sieving is weak and separation is dominated by diffusion. The gels that produced the largest d_t spacing (Team 2) also tended to have the smallest normalized band widths for the mid-MW proteins, indicating that separation efficiency and band sharpness improved together at appropriate intermediate %T/%C.

(iv) Role of %T and %C in separation efficiency

Overall, the correlation between %T/%C and separation efficiency is consistent with a pore-size-controlled sieving mechanism:

(1) Very high %T (12 %T, Team 3): pores too tight; proteins remain near the top; no usable separation or MW estimation.
(2) Intermediate %T / lower %C (10 %T, 2.9 %C, Team 2): best compromise; good spacing in d_t, resolvable ladder structure, and the most reasonable MW estimates for cytochrome, myoglobin, and hemoglobin.
(3) Intermediate %T / higher %C (10 %T, 4.0 %C, Team 1): similar range but slightly smaller effective pores, leading to diminished spacing in d_t and more overlap.
(4) Lower %T (8 %T, 3.44 %C, Team 4): improved penetration for larger proteins but still insufficiently open to resolve insulin; mid-MW proteins migrate further but do not separate dramatically better than in Team 2's gel.

These trends qualitatively match the pore-size ordering inferred from the swelling experiments: gels that swelled more (larger ξ) give deeper protein penetration but require sufficient %T/%C to maintain sieving resolution in the target MW window.

(v) Suggestions for improved separations

Improved separation in future SDS PAGE experiments would primarily require adjusting the resolving gel formulation to better match the molecular weight range of interest. Because all current gels were too tight to permit insulin migration, the most important modification would be lowering the resolving gel %T (to roughly 6-8%) and slightly reducing %C to enlarge the pore size and allow low molecular weight proteins to enter the gel matrix. For higher molecular weight species or broader MW panels, a gradient resolving gel such as 6-12 %T would provide a smoother transition in pore sizes and yield more uniform resolution across the gel. Beyond formulation changes, improvements in stacking and loading would also enhance separation. Ensuring that the stacking gel retains its low %T and correct pH would sharpen the sample plug, while careful and uniform loading would reduce distortions caused by overfilling or uneven deposition. Additional refinement could be obtained by using a molecular ladder with more closely spaced reference bands or by extending the electrophoresis run time so that ladder bands spread further down the gel, improving the accuracy of the d_t to $log_{10}(MW)$ calibration. Better control of electrophoresis conditions, especially maintaining constant voltage and minimizing temperature fluctuations, would reduce diffusion driven band broadening. Running replicate gels with slightly varied %T and %C compositions would then allow systematic optimization of the pore size regime needed to simultaneously resolve small proteins like insulin and mid range proteins such as myoglobin and hemoglobin. Collectively, these adjustments would yield sharper bands, more reliable calibration curves, and more accurate molecular weight estimations across the full protein panel.

Conclusion

This work established a direct connection between formulation, structure, and function for polyacrylamide systems used in electrophoretic separations. Kinetic monitoring of APS and TEMED initiated acrylamide polymerization showed that modest changes in monomer and initiator levels shift the group constant G within the range reported by Gelfi and Righetti, and that faster reactions produce higher molecular weight chains, as confirmed by intrinsic viscosity and Mark-Houwink analysis. Crosslinking these polymers into hydrogels with controlled %T and %C generated networks whose swelling, Flory-Rehner parameters, and pore sizes agreed with literature trends and fell in the tens of nanometers regime that governs macromolecular transport. SDS-PAGE experiments translated these structural differences into practical separation performance. Intermediate %T and relatively low %C gave the best protein resolution in the 10 to 80 kDa window, while higher %T formulations excluded even mid-sized proteins and all gels were too tight to resolve insulin.

These findings are directly relevant to several real world applications that depend on polymer network structure. Polyacrylamide remains a central material in biotechnology for protein purification and quality control, and the relationships established here between %T, %C, pore size, and sieving efficiency provide a clear framework for designing gels that can target specific molecular weight ranges in analytical laboratories. The swelling and pore size results are also applicable to hydrogel based drug delivery, where the release rate of therapeutic proteins or nucleic acids is controlled by the same permeability and network parameters measured in this study. The ability to tune pore size through modest changes in formulation is similarly important for environmental engineering uses of PAAM, such as flocculants and water treatment media, where polymer chain length and network swelling influence adsorption and transport. In soft materials engineering, the correlation between polymerization kinetics, molecular weight, and final gel architecture provides a route for designing hydrogels with programmable porosity for sensors, scaffolds, and microfluidic devices.

Taken together, the data demonstrate that rational tuning of %T, %C, and solution conditions offers a predictable handle on molecular weight, network architecture, and sieving behavior. The work also highlights how classical polymer physics links synthesis conditions to performance in systems that remain foundational across biotechnology, analytical chemistry, and engineered soft materials.

Member Contributions

M. Wang and G. Dovranova performed all data processing, curve fitting, Flory-Rehner and Mark-Houwink analysis, and figure generation. The kinetic, swelling, rheology, and SDS-PAGE experiments were carried out by all five student teams under the common course protocol, with each team responsible for a distinct subset of formulations and operating conditions. All authors contributed to interpretation of the combined dataset and to revision of the manuscript.

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Appendix

A. Brookfield-type coaxial viscometer analysis

Viscosity was measured on a Bohlin coaxial Couette viscometer equipped with LV spindles that are geometrically equivalent to the Brookfield LV #61-64 set. For each run, the instrument reported a full-scale (FS) viscosity corresponding to 100 percent torque at the selected spindle and speed. The apparent viscosity at a given rpm was obtained by scaling this value by the measured torque fraction:

$$\mu_{cP} = \left(\frac{T\%}{100}\right) \cdot \mu_{FS,cP} \,,$$

where μ_{cP} is the sample viscosity in centipoise and T% is the indicated torque percentage. The full-scale viscosity for a given spindle and speed was computed as

$$\mu_{FS,cP} = \frac{TK \times SMC \times 10000}{N_{rpm}},$$

with torque constant $TK = 0.09375$, spindle multiplier constant SMC dependent on the LV spindle geometry, and N_{rpm} the rotational speed. Viscosity values were converted to SI units using 1 cP = 10^{-6} mPa·s.

To compare data across spindles, shear rate $\dot{\gamma}$ was estimated using a shear-rate constant (SRC) derived from the Lab 4 Brookfield RV6 reference geometry:

$$\dot{\gamma} \approx SRC \cdot N_{rpm}, \qquad SRC \approx SRC_{RV6} \cdot \frac{D_{spindle}}{D_{RV6}},$$

where $SRC_{RV6} = 0.238 \frac{1}{s \cdot rpm}$ and $D_{RV6} = 14.6\, mm$, and $D_{spindle}$ is the diameter of the LV spindle. Low-torque points below roughly ten percent of full scale were flagged as lower confidence because of poorer signal-to-noise. The LV spindle mapping and constants used for the FS and SRC estimates are summarized in Table 2.

Table 1. *LV spindle mapping and constants for FS method and SRC estimate.*

LV Spindle #	Brookfield Code	Geometry	Length (mm)	Diameter (mm)	SMC	Estimated SRC ($\frac{1}{s \cdot rpm}$)
1	61	Cylindrical	65.1	18.84	6.4	0.307
2	62	Disc	6.86	18.72	32	0.305
3	63	Disc	-	12.60	128	0.205
4	64	(disc)	-	-	640	-

*Note: The diameter was not provided for LV #4 in the TA note. To place #4 data on the common shear-rate axis, we assumed SRC#4~SRC#3=0.205 $s^{-1}rpm^{-1}$. This affects only the x-axis placement ($\dot{\gamma}$); the viscosity values still come from the instrument FS method.